The Newton Flow of The Hyperbolic Tangent Of One Plus t Squared

by Stephen Crowley

February 13, 2024

The differential equation for the Newton flow $z(t)$ of $f(t)$ is given by

$$\dot{z}(t) = -\frac{f(z(t))}{\frac{d}{dt} f(z(t))} = -\frac{f(z(t))}{\dot{f}(z(t))} \tag{1}$$

If

$$f(t) = S(t) = \tanh\left(\ln\left(1+t^2\right)\right) \tag{2}$$

and we let

$$\begin{aligned} b(a) &= (1+a^2)^2 + 1 \\ &= a^4 + 2a^2 + 2 \end{aligned} \tag{3}$$

then define

$$\begin{aligned} g(t,a) &= e^t - 1 + \frac{2}{b(a)} \\ &= e^t - 1 + \frac{2}{(1+a^2)^2+1} \end{aligned} \tag{4}$$

and

$$\begin{aligned} h(t,a) &= \sqrt{e^{2t} - \frac{a^4\left(2+a^2\right)^2}{b(a)^2}} \\ &= \sqrt{e^{2t} - \frac{\left((1+a^2)^2-1\right)^2}{\left((1+a^2)^2+1\right)^2}} \end{aligned} \tag{5}$$

then there are 4 solutions of $z(t)$ for the Newton flow given by

$$\begin{aligned} z(t,a) &= \pm\sqrt{\pm\frac{g(t,a)+h(t,a)}{g(t,a)}} \\ &= \pm\sqrt{\pm\frac{e^t - 1 + \frac{2}{(1+a^2)^2+1} + \sqrt{e^{2t} - \frac{\left((1+a^2)^2-1\right)^2}{\left((1+a^2)^2+1\right)^2}}}{e^t - 1 + \frac{2}{(1+a^2)^2+1}}} \end{aligned} \tag{6}$$

which satisfy

$$\lim_{t\to\infty} z(t,a) = \{0, \pm i\sqrt{2}\} \forall a \in \mathbb{C} \tag{7}$$

where by design, definition, and necessity

$$S\left(\lim_{t\to\infty} z(t,a)\right) = 0 \forall a \in \mathbb{C} \tag{8}$$

1 Newton Flows Are Infinitesimal Generators of Semigroups of Holomorphic Mappings

The Newton flow corresponding to the solution of the continuous Newton's method can be considered as an infinitesimal generator of a semigroup of holomorphic mappings.

To understand this, we need to briefly discuss the continuous Newton's method and the concept of semigroups of holomorphic mappings.

Continuous Newton's method is a generalization of the classical Newton's method in which the iterations are replaced by a continuous flow. For a given holomorphic function $f(z)$, the continuous Newton's method is described by the following ordinary differential equation (ODE):

$$\frac{dz}{dt} = -\frac{f(z)}{f'(z)} \tag{9}$$

with an initial condition $z(0) = z_0$.

The solution of this ODE is a continuous trajectory $z(t)$ in the complex plane, which converges to the zero of the function $f(z)$ as t approaches infinity.

Definition 1

A semigroup of holomorphic mappings is a family of mappings $T(t): D \to D$, where D is an open subset of the complex plane and t belongs to a certain interval $[0, t_{\max})$, with the following properties:

1. *$T(0)$ is the identity map on D.*
2. *For all $s, t \geq 0$, if $s + t < t_{\max}$, then*

$$T(s) \circ T(t) = T(s+t) \tag{10}$$

where $\circ$ denotes the composition of mappings.

Definition 2

An infinitesimal generator A of a semigroup of holomorphic mappings $T(t)$ is a linear operator that satisfies the following condition:

$$\lim_{t\to 0}\frac{T(t)\,z(t)-z(t)}{t}=A\,z(t)\forall z(t)\in D$$

Now, consider the Newton flow as a semigroup of holomorphic mappings $T(t)$ with an infinitesimal generator A. It can be shown that the Newton flow generated by the continuous Newton's method indeed satisfies the properties of a semigroup, and its infinitesimal generator A can be represented by the linear operator

$$-\frac{f(z)}{f'(z)}=\dot{z}(t)=-\frac{f(z(t))}{\frac{d}{dt}f(z(t))}=-\frac{f(z(t))}{\dot{f}(z(t))}$$

Therefore, the Newton flow corresponding to the solution of the continuous Newton's method can be considered as an infinitesimal generator of a semigroup of holomorphic mappings.

2 Expounding a bit more...

The fundamental idea behind considering Newton flow as an infinitesimal generator is that the evolution of the flow can be captured by the differential equation that drives the flow. We have the Newton's method expressed as an ordinary differential equation (ODE), which continuously tracks the trajectory of the points on the complex plane. This trajectory forms a semigroup under the operation of function composition.

The infinitesimal generator is essentially the velocity of the system, and it encodes the dynamics of the system. When we talk about Newton flow, the infinitesimal generator is given by the right-hand side of the ODE:

$$-\frac{f(z)}{f'(z)}=\dot{z}(t)=-\frac{f(z(t))}{\frac{d}{dt}f(z(t))}=-\frac{f(z(t))}{\dot{f}(z(t))}$$

Here, $f(z)$ is the function that we're trying to find roots for, and $f'(z)$ is its derivative. As time progresses, the points of the flow evolve according to the dynamics of the system. In the limit as t approaches infinity, $z(t,a)$ converges to a set of values $\{0,\pm i\sqrt{2}\}$, regardless of the initial codition $z(0)=a$. Since $S\left(\lim_{t\to\infty}z(t,a)\right)=0$, it confirms that these indeed are the roots of $S(t)$. The whole process of determining the roots of $S(t)$ by tracking the evolution of the Newton flow is driven by the infinitesimal generator, which is $-\frac{f(z)}{f'(z)}$ for the continuous Newton's method.

The connection between the infinitesimal generator and the semigroup properties of the flow is the fundamental principle that makes this method so powerful for root-finding.